MID SEM – I (PHIC 101, **AIML, AIDS, MV, CSE, IT, M&C, IIOT**) TIME: **3:00 PM – 3:50 PM**

B.Tech (**1st Semester**) Time Allowed: **50 min** Max. Marks: **15**

NOTE: Attempt any **Three questions**. Extra ANSWER SHEETS will **NOT** be supplied.

---

(1) Discuss following processes in LASERs with neat relevant diagrams, (i) stimulated absorption (ii) stimulated emission. Discuss in brief why two level systems is not suitable for obtaining LASER? **[2+2+1]**

(2) Define the coherence length? Calculate the coherence length of LASER beams having band width 1000 Hz and 1.5 GHz. Which one is suitable for producing interference pattern and why? **[2+2+1]**

(3) Explain the light propagation mechanisms in step-index and graded–index optical fiber. Tabulate advantages and disadvantages of step-index and graded–index optical fiber. Is optical fiber bending by 60° allowed or not, justify? **[2+2+1]**

(4) Write Lorentz transformation equations. Derive the expression for length contraction. Can you write Lorentz transformation equations for photon in free space or not, justify your answer? **[1+3+1]**